

DAY16-WIRESHARK

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THEORY

1. What is Wireshark?

Wireshark is a network protocol analyzer used to capture and monitor network traffic

2. What is packet capture?

It is the process of capturing packets to analyze them.

3. What is a capture interface?

It is the network interface from which wireshark captures the packets like ethernet or wifi.

4. Name the three main Wireshark panes.

Packet lists , packet details and packet bytes

5. What is the difference between source and destination IP?

Source ip is the ip of the device that sent the packet while destination ip is for the device receiving it

6. What does the Protocol column show?

It shows the name of the protocol used for a packet like tcp,udp

7. What is a Wireshark display filter?

It is like a search panel to filter the captured traffic for easier monitoring and analyzing

8. Which filter shows ICMP traffic?

ICMP

9. Which filter shows DNS traffic?

DNS

10. Which filter shows TCP traffic?

TCP

11. What does tcp.port == 443 display?

It shows tcp packets using source or destination ip addresses that has port number 443 which is commonly used for https

12. What is DNS?

It is domain name system protocol used to translate human readable domain names to ip addresses that the computer can understand

13. What is an ICMP Echo Request?

It is a message used to test the reachability of a device commonly used by ping

14. What is an ICMP Echo Reply?

It is the response from the destination device sent to it the echo request

15. What is the TCP three-way handshake?

Syn , syn ack , ack it is the process used to establish a tcp connection

16. What are SYN, SYN-ACK and ACK?

It is the three way handshake that is used to establish a tcp connection

17. Why is HTTPS content normally not visible as plain text?

Because it is usually encrypted by tls so the content is not normally readable from packets

18. What is a PCAP/PCAPNG file?

A pcap file is a file saved in it the captured network packets

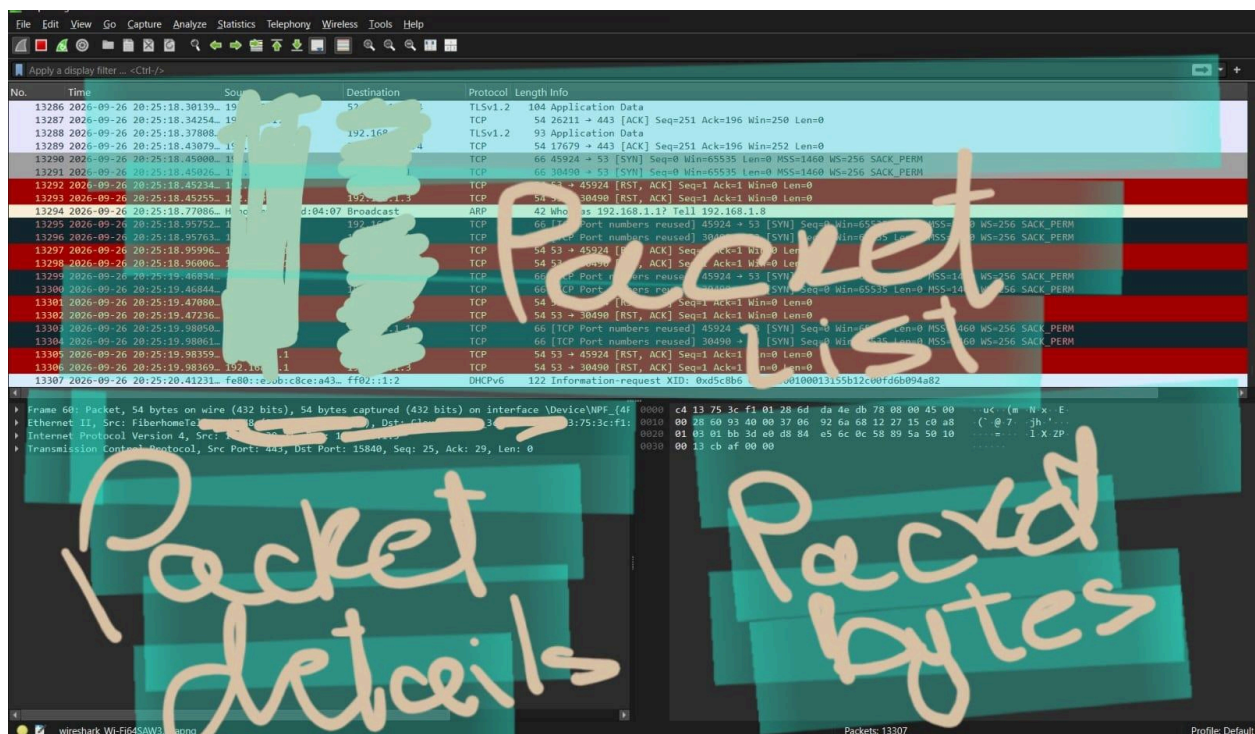
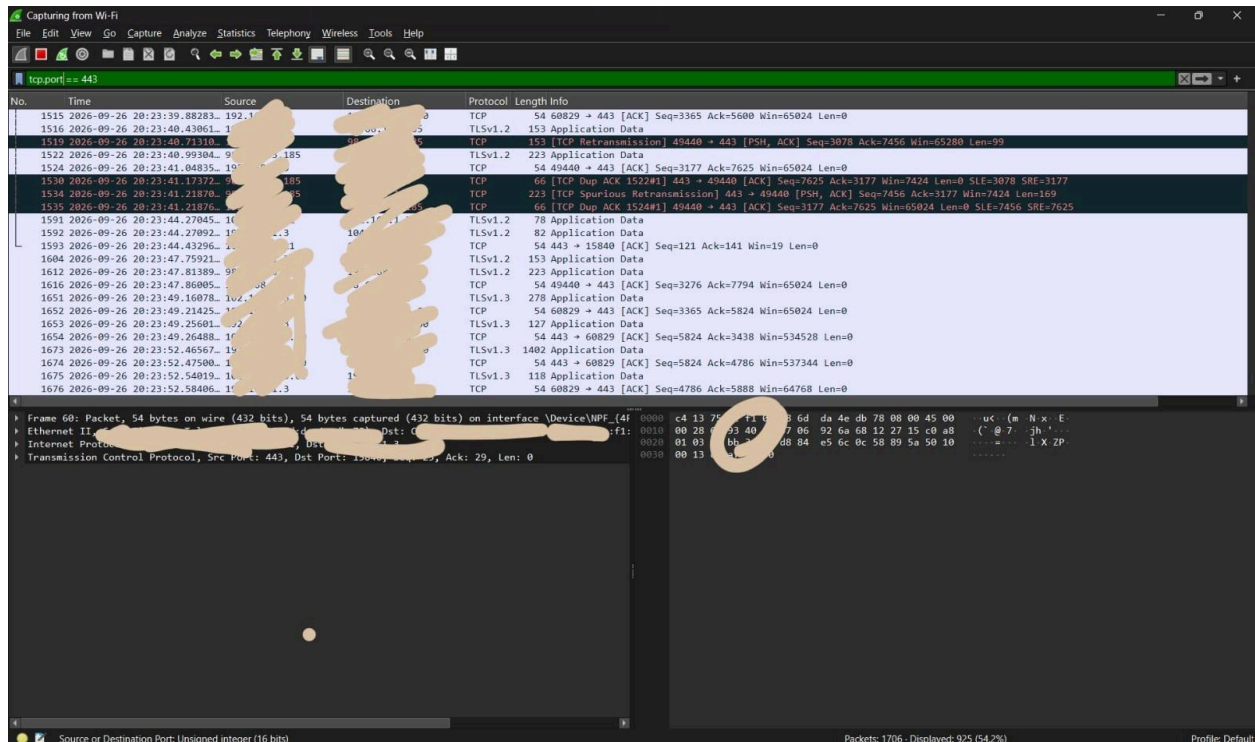
19. Why is Wireshark useful in incident response?

It helps security analysts investigate suspicious network activity, identify communication between devices, and find evidence of potential attacks.

20. Why must packet capture be performed only on authorized traffic?

Because network packets can contain private or sensitive information, and capturing traffic without authorization can violate privacy, security policies, or laws.

PRACTICAL SCREENSHOTS



day16-lab-pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter: ip.addr == 192.168.1.1

No.	Time	Source	Destination	Protocol	Length	Info
15263	2026-09-26 20:26:18.70362	192.168.1.1	192.168.1.1	TCP	54	43507 → 443 [ACK] Seq=128 Ack=521 Win=253 Len=0
15264	2026-09-26 20:26:18.708136	192.168.1.1	192.168.1.1	QUIC	1292	Protected Payload (KP0), DCID=0172097adcc21c603b71be7a94c23fc2842e9ab3
15265	2026-09-26 20:26:18.709143	192.168.1.1	192.168.1.1	QUIC	357	Protected Payload (KP0), DCID=0172097adcc21c603b71be7a94c23fc2842e9ab3
15266	2026-09-26 20:26:18.709120	192.168.1.1	192.168.1.1	QUIC	66	Protected Payload (KP0)
15267	2026-09-26 20:26:18.708216	192.168.1.1	192.168.1.1	QUIC	856	Protected Payload (KP0)
15268	2026-09-26 20:26:18.712000	192.168.1.1	192.168.1.1	QUIC	86	Protected Payload (KP0), DCID=0172097adcc21c603b71be7a94c23fc2842e9ab3
15269	2026-09-26 20:26:18.71500	192.168.1.1	192.168.1.1	UDP	1292	54743 → 443 Len=1250
15270	2026-09-26 20:26:18.71600	192.168.1.1	192.168.1.1	UDP	1292	54743 → 443 Len=1250
15271	2026-09-26 20:26:18.716300	192.168.1.1	192.168.1.1	UDP	415	54743 → 443 Len=373
15272	2026-09-26 20:26:18.717300	192.168.1.1	192.168.1.1	UDP	1286	54743 → 443 Len=1244
15273	2026-09-26 20:26:18.717780	192.168.1.1	192.168.1.1	UDP	1292	54743 → 443 Len=1250
15274	2026-09-26 20:26:18.717790	192.168.1.1	192.168.1.1	UDP	1292	54743 → 443 Len=1250
15275	2026-09-26 20:26:18.716780	192.168.1.1	192.168.1.1	UDP	1292	54743 → 443 Len=1250
15276	2026-09-26 20:26:18.716782	192.168.1.1	192.168.1.1	UDP	1292	54743 → 443 Len=1250
15277	2026-09-26 20:26:18.716783	192.168.1.1	192.168.1.1	UDP	1292	54743 → 443 Len=1250
15278	2026-09-26 20:26:18.716784	192.168.1.1	192.168.1.1	UDP	152	54743 → 443 Len=118
15279	2026-09-26 20:26:18.715284	192.168.1.1	192.168.1.1	UDP	72	443 → 54743 Len=38
15280	2026-09-26 20:26:18.715258	192.168.1.1	192.168.1.1	UDP	68	443 → 54743 Len=26
15281	2026-09-26 20:26:18.715294	192.168.1.1	192.168.1.1	UDP	74	54743 → 443 Len=32
15282	2026-09-26 20:26:18.7154468	192.168.1.1	192.168.1.1	UDP	759	443 → 54743 Len=717
15283	2026-09-26 20:26:18.715680	192.168.1.1	192.168.1.1	UDP	75	54743 → 443 Len=33
15284	2026-09-26 20:26:18.77863	192.168.1.1	192.168.1.1	ARP	42	Who has 192.168.1.1? Tell 192.168.1.8

Frame 15253: Packet, 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface \Device\NPF...
Ethernet II, Src: ..., Dst: ...
Internet Protocol Version 4, Src Port: 443, Dst Port: 54806
QUIC IETF

Ready to load or capture

Packets: 15284 - Dropped: 0 (0.0%)

Profile: Default

Capturing from Wi-Fi

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icmp

No.	Time	Source	Destination	Protocol	Length	Info
5	2026-09-26 20:22:18.47288	192.168.1.1	192.168.1.1	ICMP	74	Echo (ping) request id=0x4802, seq=1/256, ttl=64 (no response found!)
7	2026-09-26 20:22:18.57052	192.168.1.1	192.168.1.1	ICMP	74	Echo (ping) request id=0x4802, seq=2/512, ttl=64 (no response found!)
8	2026-09-26 20:22:18.67113	192.168.1.1	192.168.1.1	ICMP	74	Echo (ping) request id=0x4802, seq=3/768, ttl=64 (no response found!)
9	2026-09-26 20:22:18.78124	192.168.1.1	192.168.1.1	ICMP	74	Echo (ping) request id=0x4802, seq=4/1024, ttl=64 (no response found!)
15	2026-09-26 20:22:18.88164	192.168.1.1	192.168.1.1	ICMP	74	Echo (ping) request id=0x4802, seq=5/1280, ttl=64 (no response found!)
17	2026-09-26 20:22:18.98166	192.168.1.1	192.168.1.1	ICMP	74	Echo (ping) request id=0x4802, seq=6/1536, ttl=64 (no response found!)
18	2026-09-26 20:22:19.09382	192.168.1.1	192.168.1.1	ICMP	74	Echo (ping) request id=0x4802, seq=7/1792, ttl=64 (no response found!)
19	2026-09-26 20:22:19.19358	192.168.1.1	192.168.1.1	ICMP	74	Echo (ping) request id=0x4802, seq=8/2048, ttl=64 (no response found!)
20	2026-09-26 20:22:19.29652	192.168.1.1	192.168.1.1	ICMP	74	Echo (ping) request id=0x4802, seq=9/2304, ttl=64 (no response found!)
21	2026-09-26 20:22:19.40844	192.168.1.1	192.168.1.1	ICMP	74	Echo (ping) request id=0x4802, seq=10/2560, ttl=64 (no response found!)

Frame 5: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF...
Ethernet II, Src: ..., Dst: ...
Internet Protocol Version 4, Src: ..., Dst: ...
Internet Control Message Protocol

Internet Control Message Protocol: Protocol

Packets: 65 - Displayed: 10 (15.4%)

Profile: Default

Capturing from Wi-Fi

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dns

No.	Time	Source	Destination	Protocol	Length	Info
22	2026-09-26 20:22:19.60648...	192.168.1.3	192.168.1.1	DNS	79	Standard query 0x702a HTTPS teams.microsoft.com
23	2026-09-26 20:22:19.60671...	192.168.1.3	192.168.1.1	DNS	79	Standard query 0x54ec A teams.microsoft.com
24	2026-09-26 20:22:19.61284...	192.168.1.3	192.168.1.1	DNS	205	Standard query response 0x702a HTTPS teams.microsoft.com CNAME teams.office.com CNAME tmc-g2.tn-4.office.com SOA ns1-34.office.com
25	2026-09-26 20:22:19.61350...	192.168.1.3	192.168.1.1	DNS	260	Standard query response 0x54ec A teams.microsoft.com CNAME teams.office.com CNAME tmc-g2.tn-4.office.com A 52.123.243.14
74	2026-09-26 20:22:31.68874...	192.168.1.3	192.168.1.1	DNS	76	Standard query 0x6b26 A web.whatsapp.com
75	2026-09-26 20:22:31.70056...	192.168.1.3	192.168.1.1	DNS	129	Standard query response 0x6b26 A web.whatsapp.com CNAME mx-ds.cdn.whatsapp.net A 102.132.103.60
202	2026-09-26 20:22:32.29223...	192.168.1.3	192.168.1.1	DNS	83	Standard query 0x67ed A www.msftconnecttest.com
203	2026-09-26 20:22:32.30211...	192.168.1.3	192.168.1.1	DNS	193	Standard query response 0x67ed A www.msftconnecttest.com CNAME www.msftconnecttest.com.edgesuite.net CNAME a39.d.akamai.net
258	2026-09-26 20:22:32.87934...	192.168.1.3	192.168.1.1	DNS	82	Standard query 0x9115 A client.wms.windows.com
259	2026-09-26 20:22:32.89255...	192.168.1.3	192.168.1.1	DNS	141	Standard query response 0x9115 A client.wms.windows.com CNAME wms.notify.trafficmanager.net A 98.66.133.185
323	2026-09-26 20:22:34.09683...	192.168.1.3	192.168.1.1	DNS	76	Standard query 0x34da A mtalk.google.com
324	2026-09-26 20:22:34.09901...	192.168.1.3	192.168.1.1	DNS	131	Standard query response 0x34da A mtalk.google.com CNAME mobile-gtalk.l.google.com A 108.177.15.188
402	2026-09-26 20:22:36.88720...	192.168.1.3	192.168.1.1	DNS	76	Standard query 0x23b1 A mtalk.google.com
403	2026-09-26 20:22:36.88966...	192.168.1.3	192.168.1.1	DNS	131	Standard query response 0x23b1 A mtalk.google.com CNAME mobile-gtalk.l.google.com A 108.177.15.188
434	2026-09-26 20:22:37.61532...	192.168.1.3	192.168.1.1	DNS	81	Standard query 0x7974 A sdownload.mcafee.com
435	2026-09-26 20:22:37.62129...	192.168.1.3	192.168.1.1	DNS	250	Standard query response 0x7974 A sdownload.mcafee.com CNAME sdownload-r53.awsconsumer.mcafee.com CNAME sdownload.mcafee.com
436	2026-09-26 20:22:37.62274...	192.168.1.3	192.168.1.1	DNS	82	Standard query 0x6829 A client.wms.windows.com
438	2026-09-26 20:22:37.62853...	192.168.1.3	192.168.1.1	DNS	141	Standard query response 0x6829 A client.wms.windows.com CNAME wms.notify.trafficmanager.net A 98.66.133.185
847	2026-09-26 20:22:44.59438...	192.168.1.3	192.168.1.1	DNS	90	Standard query 0x8635 A self.events.data.microsoft.com
848	2026-09-26 20:22:44.60049...	192.168.1.3	192.168.1.1	DNS	212	Standard query response 0x8635 A self.events.data.microsoft.com CNAME self-events-data.trafficmanager.net CNAME onedscopelabs.akadns.net

Frame 74: Packet, 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface \Device\NPF... (0x...)

Ethernet II, Src: Realtek, Dst: Realtek, Length: 144

Internet Protocol Version 4, Src: 192.168.1.3, Dst: 192.168.1.1

User Datagram Protocol, Src Port: 50102, Dst Port: 53

Domain Name System (query)

0000 28 6d da 4e db 78 c4 13 75 3c f1 01 08 00 45 00 (m N x : u c : : E

0010 00 3e 53 00 00 00 00 11 64 0d c0 a8 01 03 c0 a8 :SM : : d : : :

0020 01 01 c3 b0 00 00 00 00 2a 5c 27 6b 00 00 01 : : : 5 * \k& : :

0030 00 00 00 00 00 00 03 77 65 62 08 77 68 61 74 73 : : : w eb whats

0040 61 70 70 03 63 6f 6d 00 00 01 00 01 : : : app com : : : :

Domain Name System: Protocol

Packets: 883 Displayed: 20 (2.3%)

Profile: Default

Capturing from Wi-Fi

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tcp

No.	Time	Source	Destination	Protocol	Length	Info
1375	2026-09-26 20:23:09.92562...	192.168.1.3	192.168.1.1	TCP	54	25939 → 443 [ACK] Seq=5262 Ack=271462 Win=130816 Len=0
1376	2026-09-26 20:23:09.92821...	192.168.1.3	192.168.1.1	TCP	1454	443 → 25939 [ACK] Seq=271462 Ack=5262 Win=64128 Len=1400
1377	2026-09-26 20:23:09.93087...	192.168.1.3	192.168.1.1	TCP	1454	443 → 25939 [PSH, ACK] Seq=272862 Ack=5262 Win=64128 Len=1400
1378	2026-09-26 20:23:09.93090...	192.168.1.3	192.168.1.1	TCP	54	25939 → 443 [ACK] Seq=5262 Ack=274262 Win=130816 Len=0
1379	2026-09-26 20:23:09.93375...	192.168.1.3	192.168.1.1	TCP	1454	443 → 25939 [ACK] Seq=274262 Ack=5262 Win=64128 Len=1400
1380	2026-09-26 20:23:09.93978...	192.168.1.3	192.168.1.1	TCP	1454	443 → 25939 [PSH, ACK] Seq=275662 Ack=5262 Win=64128 Len=1400
1381	2026-09-26 20:23:09.93987...	192.168.1.3	192.168.1.1	TCP	54	25939 → 443 [ACK] Seq=5262 Ack=277062 Win=130816 Len=0
1382	2026-09-26 20:23:09.94245...	192.168.1.3	192.168.1.1	TCP	1454	443 → 25939 [ACK] Seq=277062 Ack=5262 Win=64128 Len=1400
1383	2026-09-26 20:23:09.94509...	192.168.1.3	192.168.1.1	TCP	1454	443 → 25939 [PSH, ACK] Seq=278462 Ack=5262 Win=64128 Len=1400
1384	2026-09-26 20:23:09.94518...	192.168.1.3	192.168.1.1	TCP	54	25939 → 443 [ACK] Seq=5262 Ack=279862 Win=130816 Len=0
1385	2026-09-26 20:23:09.94786...	192.168.1.3	192.168.1.1	TCP	1454	443 → 25939 [ACK] Seq=279862 Ack=5262 Win=64128 Len=1400
1386	2026-09-26 20:23:09.95580...	192.168.1.3	192.168.1.1	TCP	1454	443 → 25939 [PSH, ACK] Seq=281262 Ack=5262 Win=64128 Len=1400
1387	2026-09-26 20:23:09.95592...	192.168.1.3	192.168.1.1	TCP	54	25939 → 443 [ACK] Seq=5262 Ack=282662 Win=130816 Len=0
1388	2026-09-26 20:23:09.95852...	192.168.1.3	192.168.1.1	TCP	1454	443 → 25939 [ACK] Seq=282662 Ack=5262 Win=64128 Len=1400 [TCP PDU reassembled in 1389]
1389	2026-09-26 20:23:09.96122...	192.168.1.3	192.168.1.1	TLSv1.3	1454	Application Data
1390	2026-09-26 20:23:09.96135...	192.168.1.3	192.168.1.1	TCP	54	25939 → 443 [ACK] Seq=5262 Ack=285462 Win=130816 Len=0
1391	2026-09-26 20:23:09.96383...	192.168.1.3	192.168.1.1	TLSv1.3	1389	Application Data
1392	2026-09-26 20:23:09.98614...	192.168.1.3	192.168.1.1	TLSv1.3	224	Application Data
1393	2026-09-26 20:23:10.09318...	192.168.1.3	192.168.1.1	TCP	1454	443 → 25939 [ACK] Seq=286797 Ack=5432 Win=64128 Len=1400 [TCP PDU reassembled in 1394]
1394	2026-09-26 20:23:10.09476...	192.168.1.3	192.168.1.1	TLSv1.3	1364	Application Data
1395	2026-09-26 20:23:10.09490...	192.168.1.3	192.168.1.1	TCP	54	25939 → 443 [ACK] Seq=5432 Ack=289507 Win=130816 Len=0
1396	2026-09-26 20:23:10.11362...	192.168.1.3	192.168.1.1	TCP	54	25939 → 443 [RST, ACK] Seq=5432 Ack=289507 Win=0 Len=0

Frame 63: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF... (0x...)

Ethernet II, Src: Realtek, Dst: CloudWatch, Length: 144

Internet Protocol Version 4, Src: 192.168.1.3, Dst: 192.168.1.1

Transmission Control Protocol, Src Port: 5262, Dst Port: 49683, Seq: 1, Len: 0

0000 c4 13 75 3c f1 01 70 00 00 00 00 00 00 00 00 (m N x : u c : : E

0010 00 28 00 00 00 00 00 00 7e eb 4a 7d 85 bc c0 a8 : (: 0 : : :) :

0020 01 03 00 00 00 00 00 00 00 00 00 00 00 04 : : : : : : :] : : : : P

0030 00 00 b2 50 00 00 00 00 00 00 00 00 00 00 00 : : : : : : : : : : : R : : : : : : : :

Transmission Control Protocol: Protocol

Packets: 1999 Displayed: 1004 (71.6%)

Profile: Default